॥ वसुधैव कुटुम्बकम् ॥

SYMBIOSIS INTERNATIONAL (DEEMED UNIVERSITY)

(Established under section 3 of the UGC Act 1956)

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Semester: III **Year:** AY 26-27

Subject Title: Operating Systems Lab	
EXPERIMENT No: 4	
TITLE: Arithmetic Operations	**DoP**: 03-08-26

Aim: Implement and demonstrate arithmetic operations

Learning Outcome: 1. To understand the arithmetic operations
2. To demonstrate arithmetic operations using Linux Commands

Hardware/Software:

This experiment is performed using Ubuntu OS in a virtualized environment within Windows 11. The software used is:

- VMware Workstation 17 Player (Free Virtualization Tool)
- Ubuntu 22.04.1 LTS ISO file
- VMware Tools installed on the virtual machine

Problem Statement:

A program should be written where all the following conditions should be satisfied.

1. Perform basic operations like Addition, Subtraction, Multiplication, and Division. Accept input from a user.
2. Calculate the area and perimeter of the circle. Accept radius from a user. (Assume pi=3.14)
3. Calculate employees' gross salary, where DA is 1.65% of the basic salary, and HRA is 0.30% of the basic salary. Accept the value of basic salary from a user.
4. Calculate the mean salary given by a company if the basic salary of employees A, B, C, and D are Rs.1200, 1400, 1350, and 1800 respectively.

Theory:

1. expr

The expr (expression) command is a built-in POSIX utility designed to evaluate expressions. It processes command-line arguments, evaluates the resulting expression, and prints the output to standard output.

Key Mechanics & Behavior

- **Strict Argument Separation:** Every operator and operand MUST be passed as a separate argument to the command. This means **spaces are mandatory**. Writing expr 5+5 simply prints 5+5.
- **Integer-Only:** expr does not support floating-point (decimal) numbers. Passing expr 5.5 + 2 results in an non-integer argument error.
- **Shell Escaping:** Because operators like *, <, >, &, and | have special meaning in the shell, they must be escaped with a backslash (\) or quoted when used with expr.

Syntax

expr operand1 operator operand2

2. bc

bc stands for **Basic Calculator** (or *An Arbitrary-Precision Calculator Language*). Unlike basic command-line arithmetic utilities, bc is an entire programming language with syntax similar to C.

Key Mechanics & Behavior

- **Arbitrary Precision:** Unlike standard shell arithmetic (which is limited to integers), bc handles decimals with user-defined precision.
- **Standard Input Pipe:** bc reads from stdin, making it easy to pipe output from echo, printf, or heredocs directly into it.
- **The scale Parameter:** Controls the number of decimal places used in operations (especially division). By default, scale=0 for division.

Syntax

echo "expression" | bc

1.Program:

Open | e4.sh ~/DSY OS | Save

e4.sh | ex4.sh | exp4.sh | expr4.sh

```
#! /bin/sh

echo "Enter first number (a): "
read a
echo "Enter second number (b): "
read b

val1=`expr $a + $b`
echo "Addition is $val1"

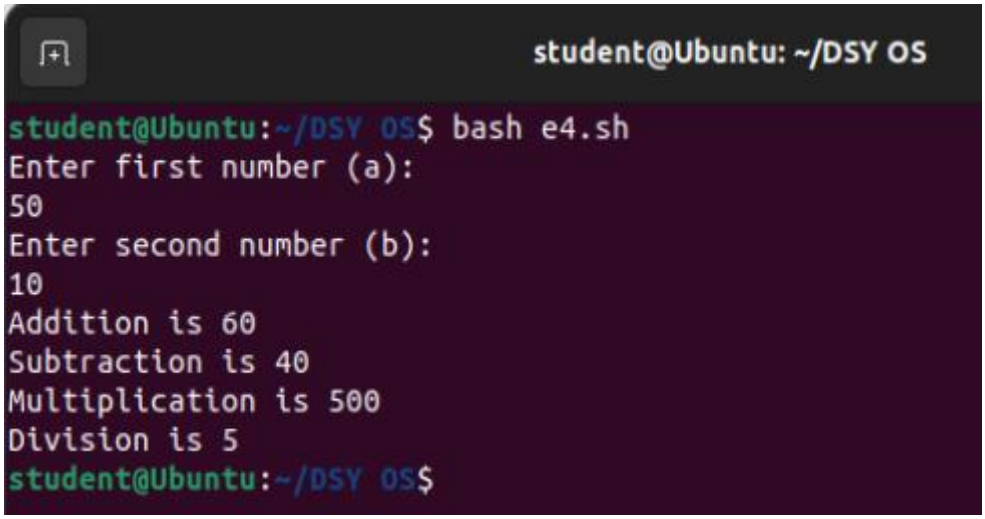
val3=`expr $a - $b`
echo "Subtraction is $val3"

val3=`expr $a \* $b`
echo "Multiplication is $val3"

val4=`expr $a / $b`
echo "Division is $val4"

```

Output Screenshots:



2.Program:

Open ex4.sh ~/DSY OS Save

e4.sh | ex4.sh | exp4.sh | expr4.sh

```
#! /bin/sh

echo "Enter radius of the circle (r): "
read r

pi=3.14

echo "Area is "
echo "$pi * $r * $r" | bc

echo "Perimeter is "
echo "2 * $pi * $r" | bc

```

Output Screenshots:

```
student@Ubuntu:~/DSY OS$ bash ex4.sh
Enter radius of the circle (r):
30
Area is
2826.00
Perimeter is
188.40
student@Ubuntu:~/DSY OS$ S
```

3.Program:

exp4.sh
~/DSY OS

e4.sh | ex4.sh | exp4.sh | expr4.sh

```
#! /bin/sh

echo "Enter Employee Basic Salary: "
read basic

da=$(echo "$basic * 0.0165" | bc)
echo "DA is $da"

hra=$(echo "$basic * 0.0030" | bc)
echo "HRA is $hra"

gross_salary=$(echo "$basic + $da + $hra" | bc)
echo "Gross Salary is $gross_salary"
```

Output Screenshots:

```
student@Ubuntu:~/DSY OS$ bash exp4.sh
Enter Employee Basic Salary:
30000
DA is 495.0000
HRA is 90.0000
Gross Salary is 30585.0000
student@Ubuntu:~/DSY OS$
```

4.Program:

e4.sh × ex4.sh × exp4.sh × expr4.sh ×

```
#! /bin/sh

echo "Enter Employee A Basic Salary: "
read A

echo "Enter Employee B Basic Salary: "
read B

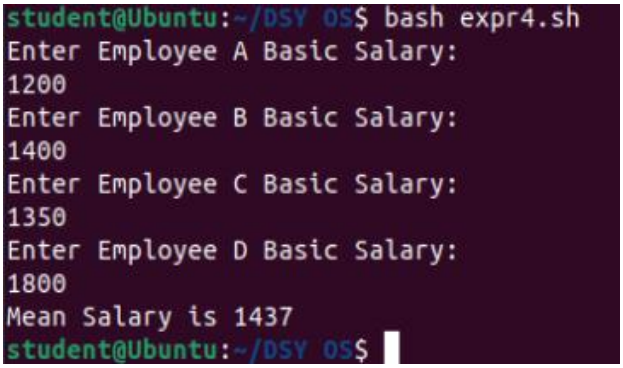
echo "Enter Employee C Basic Salary: "
read C

echo "Enter Employee D Basic Salary: "
read D

avg=$(echo "($A + $B + $C + $D) / 4" | bc)
echo "Mean Salary is $avg"

```

Output Screenshots:



Conclusion: In this experiment, arithmetic operations were successfully implemented using expr and bc shell commands in Linux. The distinction between integer arithmetic (expr) and floating-point decimal calculations (bc) was practically demonstrated through tasks involving basic operations, geometric calculations (circle area and perimeter), and employee salary evaluations.

Assessment Schemes:

Attendance	Discipline	Short oral	Correctness of Lab Report	Timely completion of Lab Report	Total marks (10)	Signature of Teacher with Date